Multiproduct Resin Reuse for Clinical and Commercial Manufacturing Part I: Resin, Column and System Performance

Rizwan Sharnez
Cleaning Validation Solutions

Krunal Mehta
Amgen

Yunjuan Wang
MedImmune

Advances in Cleaning Validation

September-October 2018

Multiproduct Resin Reuse for Clinical and Commercial Manufacturing
Part I: Resin, Column and System Performance

Rizwan Sharnez, Ph.D.	President, Cleaning Validation Solutions
Steve Doares, Ph.D.	Sr. Director, Biogen
Shane Manning	Sr. Manager, GlaxoSmithKline
Krunal Mehta, Ph.D.	Scientist, Amgen
Ekta Mahajan	Principal Engineer, Roche-Genentech
Angela To	Engineer, Hyde Engineering and Consulting
William Daniels, M.S.	Sr. Scientist, Pfizer
Judy Glynn	Sr. Manager, Pfizer
Sagar Dhamane	Senior Engineer, Amgen
Xiaona Wen	Engineer, Amgen
Yunjuan Wang, Ph.D.	Sr. Scientist, AstraZeneca
Pankaj Gour	Downstream Expert-MS&T, Novartis
Constanze Guenther, Ph.D.	Group Head Downstream Processing, Novartis
Derek Foley	Senior Engineer, Johnson & Johnson-Janssen
Ronan Hayes, M.S.	Associate Director, Johnson & Johnson-Janssen
Adam Mott	Director, Lonza
Sunil Prabhu, Ph.D.	Director, Merck & Co., Inc.
Dawn Tavalsky	Sr. Director, Sanofi
Michael Hendershot	Consultant Scientist, Eli Lilly & Co.
David Haas, Ph.D.	Associate Director, Sanofi-Genzyme
Ashley Hesslein	Associate Director, Bayer
Norbert Schuelke, Ph.D.	Director, Takeda Pharmaceuticals
Hendri Tjandra	Staff Scientist, Bayer

Keywords:
Chromatography, multiproduct resin reuse, biopharmaceutical cleaning validation.

Please address correspondence to:
Rizwan Sharnez
Cleaning Validation Solutions
3506 Vale View Lane
Mead, CO 80542, USA
rsharnez@CleaningValidationSolutions.com

Multiproduct Resin Reuse for Clinical and Commercial Manufacturing
Part I: Resin, Column and System Performance

---

1. SUMMARY

Chromatography resins used for purifying biopharmaceuticals are generally dedicated to a single product. In GMP facilities that manufacture a limited amount of any particular product, this practice can result in the resin being used for a fraction of its useful life. A methodology for extending resin reuse to multiple products is described. With this methodology, *resin, column and system performance*; *product carryover* and *cleaning effectiveness* are continually monitored to ensure that product quality is not impacted by multiproduct resin reuse (MRR).

Resin, column and system performance is evaluated in terms of *(a) system suitability parameters* such as peak-shape and transition, and height equivalent theoretical plate (HETP) data; *(b) key operating parameters* such as flow rate, inlet pressure, and pressure drop across the column; and *(c) process performance parameters* such as impurity profiles, product quality and yield (*see Part I of this series)*. Historical data are used to establish process capability limits (PCL) for these parameters. Operation within the PCLs provides assurance that column integrity and binding capacity of the resin are not impacted by MRR.

Product carryover, defined as the carryover of the *previously* processed product (*A*) in a dose of the *subsequently* processed product (*B*) ($CO_{A\rightarrow B}$), should be acceptable from a predictive patient-safety standpoint. A methodology for determining $CO_{A\rightarrow B}$ from first principles and setting acceptance limits for cleaning validation is described.

Cleaning effectiveness is evaluated by performing a blank elution run after *inter*-campaign cleaning and prior to product changeover. The acceptance limits for product carryover ($CO_{A\rightarrow B}$) are more stringent for MRR than for single product resin reuse. Thus, the *inter*-campaign cleaning process should be robust enough to consistently meet the more stringent acceptance limits for MRR. Additionally, the analytical methods should be sensitive enough to adequately quantify the concentration of the previously processed product and its degradants in the eluent.

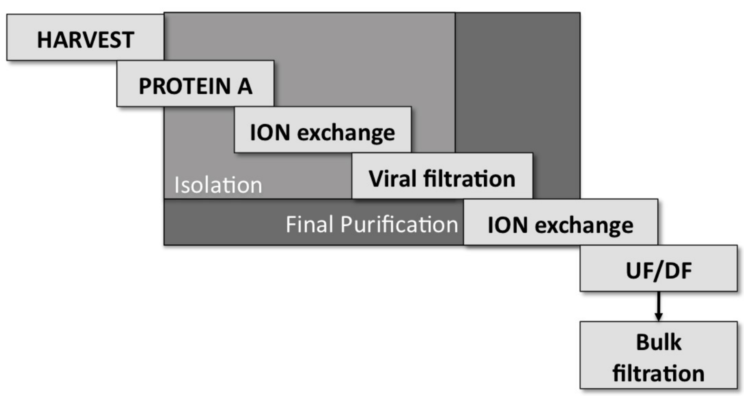
General considerations for designing small-scale chromatographic studies for process development are also described. These studies typically include process-cycling runs with multiple products followed by viral clearance studies with a panel of model viruses. The small-scale studies can be used to optimize cleaning parameters, predict resin performance and product quality, and estimate the number of multiproduct purification cycles that can be run without impacting product quality.

The proposed methodology is intended to be broadly applicable; however, it is acknowledged that alternative approaches may be more appropriate for specific scenarios.

2. INTRODUCTION

Recombinant therapeutic proteins such as monoclonal antibodies (mAb) are made in living organisms, such as Chinese Hamster Ovary (CHO) cells. The protein of interest is isolated from the cellular and media components used in its production. A typical purification process for a mAb is illustrated in Figure 1.

Figure 1: Schematic representation of purification process for a monoclonal antibody. Protein A affinity chromatography is the first column purification step in the isolation process. This is followed by ion exchange chromatography and viral filtration. The final purification step consists of a second ion exchange column, which is followed by Ultrafiltration/Diafiltration and sterile filtration.



Depending on the protein being isolated and the types of impurities present in the harvest pool, some of the purification steps in the process may be excluded or substituted with alternative steps that are more appropriate. The purification process typically consists of a few primary isolation steps followed by a final purification step. The primary isolation process begins after the protein of interest is harvested. The resulting product pool contains the protein of interest as well as cellular components (e.g. host cell proteins and DNA), media components and viruses that may be present during the production process. In the first purification step, the product pool is run through a Protein A affinity chromatography column. The purpose of this step is to remove media and cellular components, and putative viruses. Following this step, the product pool is further purified by ion exchange (IEX) chromatography. This step serves to remove additional contaminants, such as aggregates and DNA. The IEX step is followed by viral filtration and a second IEX step to remove any residual CHO proteins. The batch is then processed through an Ultrafiltration/Diafiltration (UF/DF) unit to remove small molecules, concentrate the mAb, and exchange the buffer to formulate the purified protein into its final formulation buffer. This is followed by sterile filtration and bulk-fill operations.

Chromatography is widely used to purify proteins, as it is efficient, scalable, and reproducible. However, chromatography resins used for purifying biopharmaceuticals are costly and comprise of a substantial portion of the raw material costs in biopharmaceutical manufacturing. Further, for clinical and commercial manufacturing, the resins are generally dedicated to a single product. This can result in the resin being substantially underutilized in GMP facilities that make a limited amount of any particular product.

The multiproduct resin reuse (MRR) approach described in this series can be used to increase resin utilization in all phases of clinical and commercial manufacturing. The increase in resin utilization would typically be substantially greater in clinical manufacturing where fewer lots are manufactured per product. For a clinical facility that manufactures eight new products per year, annualized savings for Protein A resin are estimated to be US $2,100,000. This estimate is based on $12,000 per litre of Protein A resin, a 30 cm column, 25 litres per column pack, six cycles per product, and resin reuse for 48 cycles; thus, the cost of the resin per product would be $12,000 per litre x 25 litres or $300,000. It follows that if the resin is re-used for eight products, or 48 cycles per year, instead of being discarded after every product, or six cycles, the annualized savings would be $2,100,000. The savings would be commensurately lower for resins that cost less, such as IEX resin.

A methodology for extending resin reuse to multiple products is described in this series. With this methodology, *resin, column and system performance*; *product carryover* and *cleaning effectiveness* are continually monitored to ensure that product quality is not impacted by MRR.

3. RESIN, COLUMN AND SYSTEM PERFORMANCE

Resin performance is evaluated in terms of the effectiveness of the resin after a given number of lots. The performance of the resin is evaluated relative to established process capability limits (PCLs) for *key product quality attributes* such as impurity profiles and yield. The PCLs are established based on historical data from full-scale operations. Operation within the PCLs provides assurance that column integrity and binding capacity of the resin are not impacted by MRR. Column performance is evaluated in terms of *system suitability parameters,* such as peak-shape and transition, and HETP data; and *key operating parameters* such as flow rate, inlet pressure, and pressure drop across the column. Control limits for these parameters are established through small-scale column-cycling studies described in Section 3.3. Operation within the control limits provides assurance that the column is operating as intended. System performance refers to the performance of the *chromatography unit operation as a whole* and includes both resin and column performance.

3.1 Leveraging full-scale data to verify stable system performance – single-product resin reuse.

Most manufacturing operations generate substantial amounts of both continuous and discrete data during downstream unit operations that can be used to verify that flow and bed integrity characteristics are operating within acceptable limits. The continuous data include operating parameters such as flow rates, inlet pressures, and pressure drops across the column. Discrete data include profiles of in-line UV, RI, and/or conductivity from the actual batch processing steps, such as load, washes, elution, strip/regeneration phases, as well as salt or solvent HETP pulses. Review or assessment of the peak shapes or buffer transitions are often part of the Manufacturing and Quality review of chromatographic unit operations. Blank elution runs described in the Appendix are typically used at some frequency – for example, at the end of a campaign – to confirm that the cleaning process remains effective.

Manufacturing operations also generate a substantial amount of process- or product-specific in-process testing data that can be used to verify resin and column performance. This includes levels of host-cell impurities such as DNA or host-cell proteins; step yields; process-related impurities, such as residual antifoam or viral inactivation detergent; and product-specific attributes, such as aggregates, half-antibody or light chain/heavy chain variants, charge isoform distribution, and glycoform distributions. Declining trends in step yield or purity, or statistically significant increase in host-cell proteins or other impurities can serve as indicators for declining resin or column performance.

All of these data may be used, in whole or in part, for what may be termed "continuous column performance monitoring" of individual columns or "resin lifetime monitoring" of individual resin lots in a given process. These lifetime monitoring programs have some utility within a short campaign but play a more significant role in longer commercial campaigns to demonstrate that a process is in control, both batch-to-batch and campaign-to-campaign. In fact, early trends in these data are usually the reason why columns are taken out of service before they have reached their maximum cycle counts as defined by small-scale column cycling studies discussed in Section 3.3.

3.2 Leveraging full-scale data to verify stable system performance – multi-product resin reuse

Conceptually, the approach described in the previous section is equally applicable to resin used across different molecules. The in-line system performance monitoring together with product quality and impurity profile trends can be used to monitor the resin and column performance when the system is used for multiple products. Similarly, key system suitability and operating parameters such as flow and pressure, peak-shape and transition, and HETP data used to verify stable system performance when the resin is dedicated to a single product (i.e. from campaign-to-campaign), can also be used to verify system performance when the resin is shared among multiple products. The assessment of peaks and transitions would be product- and buffer-specific, but the monitoring approach would be consistent with the approach described in the previous section for single product resin reuse.

As an example, consider a Protein-A resin column that is used for a five-batch campaign of mAb-A, then a ten-batch campaign of mAb-B, and then again for a seven-batch campaign of mAb-A. Although mAb-A and mAb-B may have different wash, elute and regeneration solutions, and therefore different flow and pressure characteristics, peak shapes, and peak areas, the lifetime monitoring criteria would be set specifically for each product. The criteria would additionally include an assessment of the performance of the column during the subsequent 7-batch campaign of mAb-A against the initial 5-batch campaign of that product. Well-designed control charts would be a valuable aid in these monitoring assessments. Additionally, statistically significant declines in product impurity profiles and step yields can serve as indicators for monitoring resin performance similar to the conventional approach that is used when the resin is dedicated to a single product.

3.3 Design Considerations for Small-Scale Column Cycling Studies

Small-scale column-cycling studies with representative target product load streams are used to simulate resin reuse at full scale. The studies are used during process development to establish the maximum number of cycles (N_{MAX}) for which the resin can be reused in manufacturing; the studies are also used to establish limits for process performance qualification (PPQ) runs and routine monitoring during clinical or commercial manufacturing. The process cycling runs are typically followed by small-scale viral clearance studies on the reused resin with a panel of model viruses. The results are compared to those obtained with unused resin to confirm that N_{MAX} is acceptable from a viral clearance standpoint.

Although column cycling studies have traditionally been used for single product resin reuse, they can be adapted to include multiple products and to establish N_{MAX} for the application of MRR at full scale. In order to support IND and NDA activities, development data from small-scale studies that demonstrate acceptable pre-determined column lifetime monitoring parameters – both within and across runs with different products – can provide strong support for any subsequent MRR at full scale. The following questions should be addressed when designing small-scale studies for MRR applications:

- Which classes of molecules need to be processed, e.g., monoclonal antibodies, peptibodies or fusion proteins?
- If different classes of molecules need to be processed, which resins will be reused and for which class of molecules?
- Are any of the molecules highly active or potent?
- Does one class of molecules represent a "worst-case" for a particular resin; for example, does one class of molecules tend to leave irreversible aggregates on a particular resin or require an unusually harsh buffer or cleaning step?
- How many batches will be run in each campaign? Based on this information, how many campaigns can possibly be run with the same resin?
- Do any cleaning, regeneration, storage or handling steps need to be simulated at small-scale?

The above considerations can be useful in designing small-scale studies that lay the groundwork for full-scale column lifetime monitoring. An example would be a portfolio of several monoclonal antibodies (mAb) and a couple of Fc fusion proteins that are captured using a Protein A resin: one of the mAbs (A) has a relatively low step recovery from the resin, and one of the Fc fusion proteins (B) requires a relatively high concentration of NaOH for cleaning. An experimental design that is *representative of worst-case processing conditions* might include the mAb with the relatively low step recovery (A), followed by the Fc fusion protein that is relatively difficult to clean (B), followed by a typical mAb (C). The initial cycling design could include every possible switch from one molecule to the other (i.e., A-B-C-A-C-B-A), and the subsequent cycling design could include multiple batches of the same product across a mix of campaigns (e.g., 3xA, 3xB, 5xC, 5xB, 5xA, 3xC). The monitored parameters – including mechanical performance, chromatogram and transition shapes, step yields, and in-process product and residual testing – would provide information on the feasibility of the MRR approach for these molecules and baselines for the full-scale parameters. Note that the small-scale studies may need to be repeated if a new molecule with unusual characteristics or requirements is introduced.

A sufficiently robust design of small-scale studies on a given resin can serve as a one-time "proof-of-concept" for the implementation of MRR for that resin at full scale. When using this approach, it would be advisable to perform a risk assessment each time a new product is to be introduced into the facility. This could be a part of the risk assessment that is performed as part of the technology transfer for the new molecule. The risk assessment should address whether the new product presents significant risk to resin performance. If the risk to resin performance is low, the new product could be introduced into the facility without additional studies, and it could simply be tracked under the column lifetime monitoring; however, if the new product has certain unique characteristics, additional small-scale studies with the new product may be warranted to demonstrate that it is suited for resin reuse. Examples of unique characteristics that may warrant additional small-scale studies include a very different class of molecule or a highly potent or sensitizing molecule; an abnormally high titer load, or a molecule with abnormal aggregation tendencies that could contribute to resin cleanability issues. If the results of the small-scale studies indicate that the new product presents tangible risks for MRR, the resins for that product should be dedicated.

3.4 Implementation at Full-Scale

Once the MRR approach has been justified through small-scale cycling studies, it can be implemented at manufacturing scale. The decision to implement the MRR approach for a particular resin or column is generally driven by business considerations, such as resin cost and utilization. For example, it may make sense to implement MRR for resins that are relatively expensive and underutilized, which in turn depends on whether the plant runs short campaigns of many different products or long campaigns of a few products. In the latter case, it may make more sense to dedicate resins, because underutilization of the resin is not a significant issue. The implementation of MRR for a given application would be based on closely monitoring the data discussed in Section 3.1.

The accepted approaches to monitoring and assessing these types of data for column performance over the lifetime of the resin and for campaign trends should be readily adaptable to MRR. On-line continuous data, such as flow rates and pressure, would be monitored as usual, although the control limits would clearly be adjusted as applicable to each specific product. For example, mAb-A may have an inlet pressure during load of approximately 22 psig, while for mAb-B, this inlet pressure may be approximately 18 psig at the same flow rate. Control charts would reflect the appropriate control limits for each product in the sequence of MMR runs. An important assessment of ongoing column performance would be to compare load inlet pressure for mAb-A to the control limits for mAb-A each time the column is used to purify mAb-A.

A similar approach would be followed for trended in-process data at full scale. Consider an example where the process capability limits for host-cell proteins (HCP) in a Protein-A eluate for mAb-A and mAb-B are 4 – 6 ng/mL and 28 – 32 ng/mL, respectively. A critical part of the *system* performance assessment at full-scale would be that each time mAb-A or mAb-B campaigns are run, the HCP data from each campaign would be evaluated against the corresponding HCP process capability limits established for that product (i.e. 4 – 6 ng/mL or 28 – 32 ng/mL). If the in-process HCP data for each product lie within the established PCLs for that product, it follows that column performance is not affected by any of the intervening campaigns of other products. Note that some system parameters are independent of the products processed on the column; examples include HETP and asymmetry factor. The control charts for such parameters would have a single set of control limits for all, products. These control limits would provide a holistic lifetime view of column performance across multiple products.

Cleaning cycle development for changeover between products and evaluation of cleaning effectiveness will be addressed in Parts II and III of this series. The following topics will be discussed: (a) regulatory expectations; (b) bench-scale studies for cycle development; (c) identifying target impurities; and (c) acceptance limits for cleaning validation based on maximum allowable carryover (MAC).

ACKNOWLEDGEMENTS

The authors thank the IQ Consortium for its help and support.

SYMBOLS AND ACRONYMS

AL	Acceptance limit
ADE	Acceptable daily exposure
API	Active pharmaceutical ingredient
CC	Carbon content
CO	Carryover
CIP	Clean-in-place
COP	Clean-out-of-place
DP	Drug Product
HCP	Host Cell Protein
m_B	Mass of API in a dose of Product B
M_B	Mass of API in a batch of Product B
MAC	Maximum allowable carryover of Product A in a DP batch of Product B
MRR	Multiproduct resin reuse
N	Number of doses per batch of the drug product
PCL	Process capability limit
PDE	Permitted daily exposure
Product A	Previously manufactured product
Product B	Subsequently manufactured product

RRF	Rinse recovery factor
SA	Surface area
SF	Scaling factor
SRF	Swab recovery factor
V	Volume
WFI	Water for Injection

SUBSCRIPTS

A	Product A
B	Product B
CO	Carryover
EB	Elution buffer
MRR	Multiproduct resin reuse
MPE	Multiproduct equipment
T	Total
TI	Target impurity

REFERENCES*

1. Sharnez, R., Lathia, J., Kahlenberg, D., Prabhu, S., and Dekleva, M., In Situ Monitoring of Soil Dissolution Dynamics: A Rapid and Simple Method for Determining Worst-case Soils for Cleaning Validation, *PDA Journal of Pharm. Sc. & Tech.,* Vol. 58, No. 4, p. 203-214, July-Aug 2004.
2. Canhoto, A., et al., A Novel Bench Scale Apparatus to Model and Develop Biopharmaceutical Cleaning Procedures, *J. of Validation Technology,* Vol. 11, No. 4, p. 6-14, Nov 2004.
3. Sharnez, R., To, A., Tholudur, A., Experimental Parameters for Cleaning Characterization; Part I: Dilution of Process Fluids during Cleaning, *J. of Validation Technology*, Vol 19, No.3, Sept. 2013.
4. Sharnez, R., To, A., Annapragada, R., Experimental Parameters for Cleaning Characterization; Part II: Effect of Fluid Velocity on the Kinetics of Cleaning, *Journal of Validation Technology,* Vo. 21, No. 1, 2015.
5. P. Pluta, Laboratory Studies in Cleaning Validation, *J. of Validation Technology,* Vol. 13 No. 4, 2007.
6. Verghese, G., and Kaiser, N., Cleaning Agents and Cleaning Chemistry, *Cleaning and Cleaning Validation*, Vol. 1, Chapter 7, Davis Healthcare International & PDA, 2009.
7. Verghese, G., and Thomas, J., Cleaning Agents for Biopharmaceutical Manufacturing, *Genetic Engineering News*, Vol. 23, No. 6, 2003.

8. Sharnez, R., and Klewer, L., Strategies for Developing a Robust Cleaning Process – Part II: Demonstrating Cycle Effectiveness, *American Pharmaceutical Review - Digital Edition,* Vol. 15, Issue 3, 2012.
9. Sharnez, R., and VanTrieste, M., Quality-by-Design for Cleaning Validation; *Cleaning and Cleaning Validation*, Vol 1, Chapter 6, Davis Healthcare International & PDA, 2009.
10. Canhoto, et al., A Semi-Quantitative Matrix for Selecting an Appropriate Cleaning Validation Worst-case Challenge Soiling Solution, *J. of Validation Technology,* Vol. 11, No. 4, p. 6-14, Nov 2004.
11. Sharnez, R., Leveraging Small-Scale Models to Streamline Validation, *Journal of Validation Technology,* Vol 14, No. 4, Summer 2008.
12. G. Hale, A. Drumm, P. Harrison, J. Phillips, Repeated Cleaning of Protein A Affinity Column with Sodium Hydroxide, *Journal of Immunological Methods*, Vol 171, Issue 1, p. 15-21, ISSN 0022-1759, 1994.
13. E. Mahajan, J. Werber , K. Kothary , T. Larson, One Resin, Multiple Products: A Green Approach to Purification, *Developments in Biotechnology and Bioprocessing*, ACS Symposium Series, Vol. 1125, Chapter 6, p. 87–111, 2013.
14. Sharnez, R., and To, A., Strategies for Setting Rational MAC-based Limits – Part III: Leveraging Toxicology and Cleanability Data; *Journal of Validation Technology,* Vol 17, No. 3, p. 24-28, 2011.
15. Sharnez, R., Bussiere, J., Mytych, D., Spencer, A., To, A., Tholudur, A., Acceptance Limits for Inactivated Product based on Gelatin as a Reference Impurity; *J. of Validation Technology*, Vol 19, No.1, March 2013.
16. Sharnez, R., et al., Methodology for Assessing Product Inactivation during Cleaning, *J. of Validation Technology*, Vol. 18, No. 4, p. 42-45, 2012
17. Kendrick, K., Canhoto, A., Kreuze, M., Analysis of Degradation Properties of Biopharmaceutical Active Ingredients as Caused by Various Process Cleaning Agents and Temperature; *J. of Validation Technology,* Vol 15, No. 3, p. 69, 2009.
18. Risk-Based Manufacture of Pharmaceutical Products: A Guide to Managing Risks Associated with Cross-Contamination, *International Society for Pharmaceutical Engineers (ISPE)*, Second Edition, 2017.
19. Fourman, G. L. and Mullen, M. V., Determining Cleaning Validation Acceptance Limits for Pharmaceutical Manufacturing Operations, *Pharmaceutical Technology, 17 (4), 54-60, 1993*.
20. Sharnez, R., Unpublished results.
21. Sharnez, R., To, A., Klewer, L., Strategies for Setting Rational MAC-based Limits – Part II: Application to Rinse Samples, *J. Validation Technology, Vol. 17, No. 2, p. 43-46, 2011.*

*The above references are for Parts I through III of this series

rsharnez@CleaningValidationSolutions.com
(970) 535-2130 (office)
(720) 745-2978 (cell)

Rizwan Sharnez is an independent consultant to the pharmaceutical industry. His expertise is in the areas of Cleaning and Process Development and Validation. Previously he was with BioSeparations, Merck and Amgen, where he was Scientific Director. Rizwan has 25 years' experience in the pharmaceutical industry, over 60 peer-reviewed publications, 30 professional awards and 16 patents. He holds a B. Tech (IIT), M.S. and Ph.D. in Chemical Engineering/Microbiology.